IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-267

1. Topic (12 words max)

Smart Bus System for Real-Time Passenger Management

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Public transport is one of the main services in Sri Lanka. A majority of the population relies on public transport services, such as the public bus system, to meet their daily travel needs. According to the Sri Lanka Transport Board, in 2022, approximately 1.45 million people used SLTB buses for daily travel [1]. Therefore, the public bus service plays a crucial role in the country's transportation infrastructure.

However, this service faces several challenges. Some of the key issues observed are as follows:

- **Buses do not arrive on time** – Many passengers miss their buses due to schedule inconsistencies [4].
- **Overcrowding during peak hours** – Buses become excessively crowded, making it difficult for people to reach their destinations on time [2].
- **Communication barriers with drivers and conductors** – Passengers often struggle to communicate clearly about where to stop the bus.
- **Driver and conductor behavior** – Some drivers and conductors do not treat passengers respectfully or professionally [3].

Additionally, passengers currently lack access to the following essential facilities [5]:

- The ability to check accurate bus schedules and real-time arrival information.
- Information on whether a bus is crowded or not.
- A simple and effective way to communicate with drivers or conductors.
- A reliable system to lodge complaints about service-related issues.
- An easy method to reserve seats in advance.

To address these issues, we propose the development of an **IoT-driven smart mobile application** aimed at improving the efficiency and reliability of the public transport system. Our goal is to provide intelligent solutions to the problems mentioned above, thereby enhancing the overall commuting experience for the public.

References:
[1] Sri Lanka Transport Board, "Basic Information," Ministry of Transport, 2022. [Online]. Available: https://www.transport.gov.lk/web/index.php?option=com_content&view=article&id=25&Itemid=145&lang=en#no-of-passengers-transported. [Accessed: Jun. 21, 2025]

[2]T.Amalan, "Analysis of Bus Passenger Demand and Supply," University of Moratuwa,2021. [Online]. Available: https://ijsrm.net/index.php/ijsrm/article/view/3492 [Accessed: Jun. 21, 2025].

[3]Daily Mirror Online, "Disciplinary action taken against SLTB bus conductor who harassed children in Nuwara Eliya: Bimal," Mar. 07, 2025. [Online Video]. Available: https://www.youtube.com/watch?v=A25bpJIfQok. [Accessed: Jun. 21, 2025]

[4] Auditor General's Department, "Performance Report - Sri Lanka Transport Board," 2023. [Online]. Available: http://auditorgeneral.gov.lk/web/images/audit-reportsupload/2023/Performance_Reports/SLTB/3.-English.pdf. [Accessed: Jun. 21, 2025].

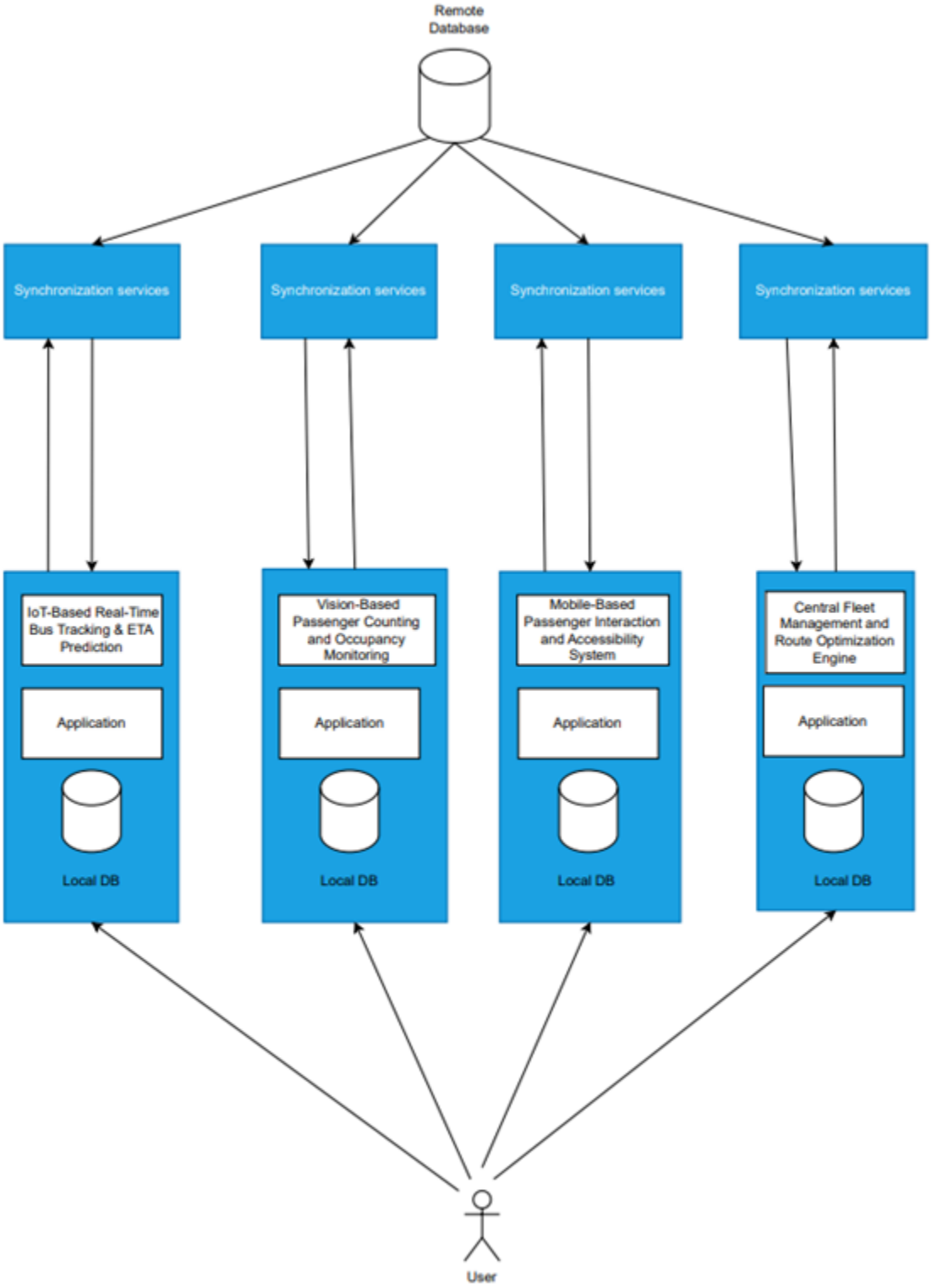
[5] Ranawana, HWMISHewage, "Factors Affecting Service Quality in Public Bus Transportation in Sri Lanka," Kothalawala Defence University,2015. [Online]. Available:http://ir.kdu.ac.lk/handle/345/1417 [Accessed: Jun. 21, 2025].

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

Our research proposes a state-of-the-art Smart Bus System designed to revolutionize public transportation through intelligent, connected, and inclusive technologies. The system is divided into four key modules, each addressing a specific challenge in modern transit and collectively creating a seamless smart mobility ecosystem.

- **IoT-Based Real-Time Bus Tracking & ETA Prediction**
  This module gathers real-time data from buses using sensors and GPS. After being processed locally, this data is synced with a database located remotely. For smooth tracking, a cloud-based LSTM model then forecasts precise ETAs by taking weather, traffic, and contextual information into account.
- **Vision-Based Passenger Counting and Occupancy**
  To identify and count passengers anonymously, the technology makes use of thermal sensors and integrated cameras. This information guides judgments on route optimization and aids in the real-time monitoring of bus occupancy.
- **Mobile-Based Passenger Interaction and Accessibility System**
  This module focuses on inclusive design for users. A mobile app with multilingual, audio, and vibration feedback allows all users including differently abled individuals to access live journey updates, request stops, and receive seat availability alerts.
- **Central Fleet Management and Route Optimization Engine**
  This part combines information from every module. It optimizes the distribution of resources by dynamically modifying routes according to demand, past usage trends, and traffic circumstances. Additionally, the engine synchronizes with multimodal services like trains or rideshares

Conceptual Diagram: -



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Developing a comprehensive Smart Bus System demands expertise across multiple technical domains and significant data integration to create an efficient public transportation solution. The project requires knowledge in IoT systems, computer vision, mobile development, and data analytics to deliver real-time tracking, passenger monitoring, and route optimization.

- **IoT and Real-Time Systems:** Essential for GPS-based bus tracking using Arduino hardware with hybrid GSM and Wi-Fi connectivity. This requires understanding edge computing, sensor integration, and real-time data transmission for uninterrupted operation across diverse environments. A low-cost, scalable deployment model supports use in developing regions.
- **Computer Vision and Deep Learning:** Vital for passenger counting and occupancy monitoring using embedded deep learning models like Tiny-YOLOv3 or MobileNet. Experience with depth cameras, thermal sensors, and privacy-preserving detection methods ensures accurate, anonymous passenger monitoring on resource-limited embedded systems.
- **Mobile App Development and HCI:** Needed for developing accessible passenger interfaces with multi-modal interactions audio, vibration, and multilingual support for differently abled users. QR code scanning, real-time UI updates based on bus capacity, and demand-responsive routing are central to the user experience.
- **Data Analytics and Fleet Management:** Key for route optimization and predictive scheduling by analyzing historical ridership patterns, traffic conditions, and passenger density. AI-driven schedule refinement and integration with train and ride-sharing services enable unified, adaptive route planning.

The system requires diverse data: real-time GPS, weather, traffic updates, passenger movement patterns, historical ridership records, and mobile user interactions. Context-aware ETA prediction with LSTM models and high-quality mapping data enables precise, demand-aware fleet management and route adjustment.

8. Objectives and Novelty

Main Objective: In Sri Lanka most people use public bus services daily. Public bus services are famous in Sri Lanka because it is very cheap. But on the other hand, there are so many Deficiencies in these bus services. So, we plan provide some solution to Resolve passenger's day to day problems when using public bus services.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Weerakoon W.M.D.P	IoT-Based Real-Time Bus Tracking & ETA Prediction	1) Design and implement GPS-based tracking system using Arduino 2) Set up cloud data pipeline for real-time bus data 3) Train LSTM model for ETA prediction using GPS, traffic, weather, and event data 4) Integrate IoT modules with GSM and Wi-Fi connectivity	The system uses a hybrid connectivity model combining GSM and Wi-Fi to ensure stable data transmission in varying environments. A context-aware LSTM model, trained on GPS, traffic, weather, and event data, enables accurate ETA predictions. The solution is low-cost and scalable, using Arduino-based edge processing, ideal for deployment in developing regions.

Pathiraja H.P.M.O.N - IT22230492	To develop a smart fleet management and route optimization engine using data analytics and optimization	1) Analyze real-time and historical data to optimize bus routes and schedules. 2) Integrate other transport modes (train etc.) for unified route planning. 3) Continuously refine schedules using AI feedback.	The system makes real-time changes to bus routes based on how many passengers are on board and the current traffic situation, instead of following a fixed timetable. It uses AI to predict and improve schedules by learning from both past and live data through a continuous feedback process. A special feature of this system is its ability to connect with other transport options like trains and ride-sharing services, helping passengers enjoy a smoother and faster journey.
Chandrasekara C.M.P.V IT22286246	To develop a mobile-based passenger interaction and accessibility system using inclusive design and real-time communication	1) Design a mobile app for real-time passenger interaction. 2) Add accessibility features like audio, vibration, and multilingual support. 3) Connect the app with the smart bus system for live data exchange. 4) Enable features like stop requests and seat availability updates.	Provides audio, vibration, and multilingual support for differently abled users. Allows passengers to request stops, check seat availability, and get live updates. Suggests smarter travel options based on real-time occupancy data.
Abeysekara W.R.G.M IT22271150	To develop a Vision-Based Passenger Counting and Occupancy Monitoring	1) Integrate real-time crowd monitoring on the bus using onboard IoT devices. 2) Transmit crowd data to the mobile application to provide	The system employs a non-GPU deep learning implementation using lightweight models such as Tiny-YOLOv3 or MobileNet, specifically optimized for performance on embedded systems with limited

		passengers with live occupancy updates. 3) Utilize thermal sensors for highly accurate passenger detection and crowd estimation.	computational resources. To ensure high detection accuracy, a hybrid sensing strategy is adopted, combining input from binocular depth cameras and thermal sensors to effectively estimate crowd density under varying conditions. Additionally, the solution incorporates privacy-preserving computer vision techniques, ensuring that all passenger detection is performed in an anonymized manner, without capturing or processing any personally identifiable information, thereby maintaining user privacy and complying with data protection standards.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Weerakoon W.M.D.P. - IT22131560	IoT-based real-time bus tracking system with ETA prediction component focuses on implementing an IoT-based real-time bus tracking system with ETA prediction. This aligns with the IT specialization through the use of embedded systems programming (Arduino), network communication (GSM/Wi-Fi), cloud integration, and machine learning (LSTM model). It combines hardware-software integration and data analytics, which are core IT skills.
Pathiraja H.P.M.O.N - IT22230492	Smart fleet management and route optimization engine component directly relates to the IT specialization through big data analysis, AI-based decision-making, and real-time system integration. It involves analyzing historical and real-time data to dynamically adjust routes, a task that heavily relies on IT skills in data science and intelligent systems.
Chandrasekara C.M.P.V IT22286246	Mobile-based passenger interaction and accessibility system component reflects core IT skills in mobile app development, human-computer interaction (HCI), and real-time data integration. The use of multilingual, audio, and vibration feedback supports inclusive IT system design, aligning closely with the principles of software development and accessibility engineering in IT.
Abeysekara W.R.G.M- IT22271150	Vision-based passenger counting and occupancy monitoring system component aligns with the IT specialization through the use of computer vision, embedded deep learning (Tiny-YOLOv3, MobileNet), and privacy-preserving data processing. It integrates software engineering, AI, and sensor data processing—all key areas in IT.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Nalin	Amarasena	
Co-Supervisor	Ms.	Uthpala	Samarakoon	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

<u>This part is to be filled by the Topic Screening Staff members.</u>

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	X	No	

b) Does the proposed topic exhibit novelty?

Yes	X	No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes	X	No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	X	No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Highly recommended

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	✓
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Accepted as it is

Staff Member's Name	Signature
Nelum Amarasena	
Uthpala Samarakoon	

***Important**:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.